

Vardaan Mehandiratta

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Education

University of Waterloo

Bachelor of Applied Science in Computer Engineering

Waterloo, ON

2025 - 2030

- Scholarships: President's Scholarship of Distinction; David Johnston and International Student Entrance Scholarships

Experience

Full Stack Developer

APi Group (Habtech Division)

Mississauga, ON

Apr 2026 - Present

- Led development of the admin and technician field reporting apps, building a **Fabric.js**-based map workflow for task assignment and technician visibility across **1,000+ site assets**.
- Developed a commission wizard that enabled non-technical users to create commissions, assign tasks, and dispatch technicians on site using **Next.js, C#/.NET, AWS, and DynamoDB**, reducing manual coordination by **90%**.
- Built a company-wide payroll and timesheet system with multi-stage approval workflows, **Azure** authentication, and batch processing, replacing spreadsheet-based operations for **1,000+ employees** across Canada.
- Entrusted as one of the first co-op students to test and modernize a thirty-year-old **C++** security platform, tracing dependencies across a large legacy codebase and developing a protocol layer for integration with a modern **C#** interface.

Software Engineering Intern

Bharat Denim

Remote, Canada

May 2025 - Aug 2025

- Delivered backend features for an inventory management system serving **10,000+ customers**, improving reliability across inventory and order workflows.
- Optimized **MySQL** and **PostgreSQL** schemas through normalization, index additions, and stricter constraints, reducing daily processing time by **15%** and improving scalability.

Projects

Memory-Driven Netflix Recommendation Engine | *React, TypeScript, Tailwind CSS, Supabase, Backboard API*

- Built a conversational Netflix recommender with persistent **Backboard API** memory and feedback-based ranking, reducing content search time by **48%**.
- Designed a **Supabase**-backed preference loop that retained feedback, prevented duplicates, and re-ranked suggestions with **92% recommendation accuracy**.
- Cut recommendation latency by **85%** (2s to <300ms) and redundant LLM calls by **70%** using **Redis** caching, TTL expiry, and per-user rate limits.

CXR-Triage (CXC AI Hackathon) | *Next.js, TypeScript, Python, PyTorch, Supabase*

- Built a full-stack medical triage system in a **36-hour sprint**, connecting a Python inference engine to a Next.js interface that prioritized critical X-rays for radiologists.
- Trained a custom **CNN** in PyTorch using augmentation and hyperparameter tuning, achieving **80% classification accuracy**.
- Secured patient data with PostgreSQL row-level authorization and an append-only audit trail.

AI Interview (McHacks 13) | *React, TypeScript, Tailwind CSS, Gemini API, ElevenLabs, Supabase*

- Built a real-time interview simulator tracking **11+ performance metrics** across technical and behavioural responses.
- Integrated **Gemini Pro** and **ElevenLabs** for automated assessments and speech transcription, achieving an **88% success rate**.

Skills & Interests

Technical Skills

Languages: C++, C#, Python, TypeScript, JavaScript, C, Go, SQL

Frontend: React, Next.js, Tailwind CSS

Backend: .NET, Node.js, Express.js, FastAPI

Cloud & Infra: AWS, Azure, Docker, Redis, Vercel, CI/CD

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, Supabase

Tools & ML: Git, Jest, Postman, Visual Studio, PyTorch, pandas, NumPy

Beyond Code

Football/Soccer: Playing and watching

Pool: Casual and competitive

Singing: Covers and performances

Gaming: Competitive and story-driven

Building: Hackathons and side projects

Ambition: Building toward a future startup